Bijections, Inverses, and Cardinality

MTH 317H — Week 6

Name: _____

1 From functions to bijections

A function $f\colon X \to Y$ assigns exactly one element $f(x) \in Y$ to each $x \in X$. The set X is its *domain*, Y its *codomain*, and $f(X) = \{f(x) : x \in X\}$ its *image*.

- f is **injective** if, for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. Equivalently, distinct inputs have distinct outputs.
- f is **surjective** if, for every $y \in Y$, there is an $x \in X$ such that $f(x) = y$. Equivalently, $f(X) = Y$.
- f is **bijective** if it is both injective and surjective. Equivalently, each $y \in Y$ has exactly one preimage in X.

Surjectivity gives *existence* of a preimage: This means, for every $y \in Y$, there is at least one **preimage** $x \in X$ with $f(x) = y$.

Injectivity gives *uniqueness* when a preimage exists: i.e. if $f(x_1) = f(x_2)$, then $x_1 = x_2$. This means that if a preimage exists, it is unique.

Warm-up

Let $A = \{1, 2, 3\}$ and $B = \{\text{dog}, \text{cat}\}$. How many functions $A \to B$ are there? How many are injective, surjective, or bijective?

Each of the three inputs has two possible outputs, independently, so there are $2^3 = 8$ functions. None is injective: three inputs cannot have three distinct outputs in a two-element set. Exactly two functions are constant; the remaining six use both outputs and are surjective. None is bijective.

For example, the rule $1 \mapsto \text{dog}$, $2 \mapsto \text{dog}$, $3 \mapsto \text{cat}$ is surjective but not injective. In the other direction, $\text{dog} \mapsto 2$, $\text{cat} \mapsto 1$ defines an injective function $B \to A$, but it is not surjective because 3 is missed.

If instead $C = \{\text{dog}, \text{cat}, \text{bunny}\}$, the assignments $1 \mapsto \text{cat}$, $2 \mapsto \text{bunny}$, $3 \mapsto \text{dog}$ define a bijection $A \to C$.

Cardinality

We write $|X| = |Y|$ when there is a bijection between X and Y; we say that they have the *same cardinality*. For finite sets, this means that they have the same number of elements. The same definition also lets us compare infinite sets.

2 Inverse functions

The identity function on X is $\mathrm{id}_X \colon X \to X$, $\mathrm{id}_X(x) = x$. If $f \colon X \to Y$ and $h \colon Y \to X$, then

$$(h \circ f)(x) = h(f(x)), \qquad (f \circ h)(y) = f(h(y)).$$

The function h is an **inverse** of f if

$$h \circ f = \mathrm{id}_X \qquad \text{and} \qquad f \circ h = \mathrm{id}_Y .$$

We write $h = f^{-1}$. This notation means an inverse function, *not* the reciprocal $1/f(x)$.

Theorem: a function is bijective if and only if it has an inverse

Proof. First suppose h is an inverse of f. If $f(x_1) = f(x_2)$, applying h gives $x_1 = x_2$, so f is injective. Given any $y \in Y$, the element $x = h(y) \in X$ satisfies $f(x) = f(h(y)) = y$, so f is surjective.

Conversely, suppose f is bijective. For each $y \in Y$, surjectivity provides a preimage and injectivity makes it unique. Define $h(y)$ to be that unique preimage. Then $f(h(y)) = y$. Also x is the unique preimage of $f(x)$, so $h(f(x)) = x$. Thus h is an inverse of f. □

The inverse is unique. If h and k are both inverses, then for every $y \in Y$,

$$h(y) = h(f(k(y))) = k(y).$$

Thus writing *the* inverse is justified.

Example: a linear function

Let $f \colon \mathbb{R} \to \mathbb{R}$ be $f(x) = 3x - 2$. Solving $y = 3x - 2$ gives $x = (y + 2)/3$. Define $h \colon \mathbb{R} \to \mathbb{R}$ by $h(y) = (y + 2)/3$. Then

$$h(f(x)) = \frac{3x - 2 + 2}{3} = x, \qquad f(h(y)) = 3\frac{y + 2}{3} - 2 = y.$$

Both compositions are identities, so f is bijective and $f^{-1} = h$.

Why check both compositions?

The equation $h \circ f = \mathrm{id}_X$ alone implies injectivity of f; the equation $f \circ h = \mathrm{id}_Y$ alone implies surjectivity of f. Neither identity alone is the definition of a two-sided inverse.

For instance, let $f \colon [0, \infty) \to \mathbb{R}$ be $f(x) = x$ and let $h \colon \mathbb{R} \to [0, \infty)$ be $h(y) = |y|$. Then $h(f(x)) = x$ for every $x \geq 0$, but $f(h(-1)) = 1 \neq -1$. The function f is not surjective, so it has no inverse with domain all of $\mathbb{R}$.

3 Bijections between infinite sets

Example: integers and even integers

Write $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$. The function

$$E\colon \mathbb{Z} \to 2\mathbb{Z}, \qquad E(n) = 2n$$

is injective, since $2n = 2m$ implies $n = m$. It is surjective because each $y \in 2\mathbb{Z}$ has the form $y = 2k = E(k)$ for an integer k. Its inverse is $y \mapsto y/2$, and therefore $|\mathbb{Z}| = |2\mathbb{Z}|$. An infinite set can have the same cardinality as a proper subset.

Example: positive integers and all integers

In these notes, $\mathbb{N} = \{1, 2, 3, \ldots\}$. Define

$$G\colon \mathbb{N} \to \mathbb{Z}, \qquad G(n) = \begin{cases} n/2, & n \text{ even}, \\ -(n-1)/2, & n \text{ odd}. \end{cases}$$

The successive values are $0, 1, -1, 2, -2, 3, -3, \ldots$. More precisely, the inverse is

$$J\colon \mathbb{Z} \to \mathbb{N}, \qquad J(z) = \begin{cases} 2z, & z > 0, \\ 1 - 2z, & z \le 0. \end{cases}$$

If $z > 0$, then $J(z) = 2z$ is positive and even, so $G(J(z)) = z$. If $z \le 0$, then $J(z) = 1 - 2z$ is positive and odd, so $G(J(z)) = -((1 - 2z) - 1)/2 = z$. Conversely, if $n = 2k$ with $k \ge 1$, then $J(G(n)) = J(k) = 2k = n$. If $n = 2k + 1$ with $k \ge 0$, then $J(G(n)) = J(-k) = 1 + 2k = n$. Hence G is bijective and $|\mathbb{N}| = |\mathbb{Z}|$.

Example: intervals

Let $a < b$. The function $L\colon (0,1) \to (a,b)$, $L(t) = a + (b-a)t$, has inverse

$$L^{-1}\colon (a,b) \to (0,1), \qquad L^{-1}(y) = \frac{y-a}{b-a}.$$

Indeed, $b - a > 0$ shows that the stated intervals are respected, and

$$L^{-1}(L(t)) = t, \qquad L(L^{-1}(y)) = y.$$

Thus $|(0,1)| = |(a,b)|$, regardless of the positive length $b - a$.

Not all infinite sets have the same cardinality

The handwritten notes also mention Cantor's theorem: there is no bijection between $\mathbb{N}$ and $\mathbb{R}$. This fact is stated here without proof; it is proved by a diagonal argument. Being infinite does not by itself imply that two sets have equal cardinality.

4 Composition and writing complete proofs

For $f\colon X \to Y$ and $g\colon Y \to Z$, the composition is

$$g \circ f\colon X \to Z, \qquad (g \circ f)(x) = g(f(x)).$$

Apply f first, then g. The domains must allow these successive applications.

Theorem: a composition of bijections is a bijection

Suppose f and g are bijective. The function $h = f^{-1} \circ g^{-1}\colon Z \to X$ satisfies

$$h((g \circ f)(x)) = f^{-1}(g^{-1}(g(f(x)))) = x,$$
$$(g \circ f)(h(z)) = g(f(f^{-1}(g^{-1}(z)))) = z.$$

Thus $g \circ f$ has an inverse, so it is bijective, and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

The order reverses because we undo the last function first.

A guide to the different proof tasks

- **To check that a rule defines a function:** prove that every input in the domain has exactly one output and that the output lies in the stated codomain.
- **To prove injectivity:** take arbitrary inputs x_1, x_2, assume $f(x_1) = f(x_2)$, and derive $x_1 = x_2$.
- **To disprove injectivity:** give two distinct inputs in the domain with the same output.
- **To prove surjectivity:** begin with an arbitrary target y in the codomain. Find an input x in the domain and verify $f(x) = y$.
- **To disprove surjectivity:** find a target in the codomain and show that no input maps to it.
- **To find and verify an inverse:** solve $y = f(x)$ for x, state the inverse's domain and codomain, check that its values belong to that codomain, and verify both compositions.

An important distinction: formula, domain, and codomain

The function $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, is neither injective nor surjective: $f(1) = f(-1)$, and no input maps to -1. The function $q\colon [0, \infty) \to [0, \infty)$ with the same formula *is* bijective, with inverse $q^{-1}(y) = \sqrt{y}$. For $x, y \geq 0$, $\sqrt{x^2} = x$ and $(\sqrt{y})^2 = y$. Changing the domain or codomain can change whether an inverse exists.

Before finishing a proof, check: Did I use arbitrary inputs or targets where required? Did I check the domain restrictions? If I claimed an inverse, did I verify both identities?